

LECTURE 06 · REVIEW NOTES

FX Forwards & Futures

How forward contracts and futures eliminate FX uncertainty, when to use each instrument, and the mechanics of covered interest arbitrage.

About these notes

These notes accompany Lecture 06 of FINA 405 · International Financial Management (University of San Diego, Knauss School of Business, Fall 2026). Content is presented in textbook style with all calculations independently verified. In-class exercises are described by topic and structure only.

1. From IRP to Forward Contracts

Lecture 5 established the Interest Rate Parity (IRP) formula as the no-arbitrage relationship between spot rates, forward rates, and interest differentials:

$$F = S \times (1 + r_{\text{quote}} \times T) / (1 + r_{\text{base}} \times T)$$

This formula is not merely a theoretical benchmark — it is the fair market price of a forward contract. A forward contract is an over-the-counter (OTC) agreement between two parties to exchange a specified amount of one currency for another at a pre-agreed rate on a future settlement date.

- No upfront premium is required — the entire cost is embedded in the forward rate.
- The contract is customizable: any amount, any currency pair, any maturity date.
- FX risk is fully eliminated for the hedging party.
- Unlike exchange-traded instruments, OTC forwards have no secondary market liquidity.

If the market forward deviates from the IRP-implied rate, covered interest arbitrage (CIA) immediately drives the market rate back. In liquid FX markets, this self-correction takes seconds.

2. Covered Interest Arbitrage: Mechanics and Quantification

Covered interest arbitrage exploits a discrepancy between the market forward rate and the IRP no-arbitrage rate. The profit is locked in at the time of trade with no residual market risk, because the forward leg hedges the foreign-currency exposure.

General CIA decision rule when $F_{\text{market}} > F_{\text{IRP}}$ (base currency overpriced in the forward market):

- Borrow in the quote currency (USD) at r_{quote} .
- Convert to base currency at the spot rate S .
- Invest base currency at r_{base} .
- Sell base currency forward at the inflated F_{market} .

- At maturity: receive USD proceeds from the forward sale, repay USD loan, keep the spread.

WORKED EXAMPLE 1**CIA Profit Calculation: GBP/USD (Review of Lecture 5 Exercise)**

Step	Action & Calculation	Amount
Given	GBP/USD spot = 1.2500; $r_{\text{USD}} = 5\%$; $r_{\text{GBP}} = 2\%$; $T = 1\text{yr}$; $F_{\text{mkt}} = 1.3000$	—
F_{IRP}	$1.2500 \times (1.05 / 1.02)$	= 1.2868
Compare	$F_{\text{mkt}} = 1.3000 > F_{\text{IRP}} = 1.2868 \rightarrow$ GBP overpriced in forward	—
① Borrow USD	\$1,000,000 at 5%; repayment at $T=1$	Owe \$1,050,000
② Convert to GBP	$\$1,000,000 \div 1.2500$ at spot	£800,000
③ Invest GBP	£800,000 at 2% for 1 year	£816,000
④ Sell GBP fwd	$\text{£}816,000 \times F_{\text{mkt}} = \text{£}816,000 \times 1.3000$	\$1,060,800
Profit	$\$1,060,800 - \$1,050,000$	+\$10,800 risk-free

The profit arises because GBP is overpriced in the forward market ($F_{\text{mkt}} = 1.3000 > F_{\text{IRP}} = 1.2868$). Selling GBP forward at the inflated rate while earning only $r_{\text{GBP}} = 2\%$ on the invested principal yields a risk-free spread.

IN-CLASS ACTIVITY · CIA Group Exercise

Three student groups each worked a CIA exercise with a different currency pair: Group 1 used EUR/USD (spot 1.0800, $r_{\text{USD}} 5\%$, $r_{\text{EUR}} 1.5\%$, $F_{\text{mkt}} 1.1400$); Group 2 used AUD/USD (spot 0.6500, $r_{\text{USD}} 5\%$, $r_{\text{AUD}} 3.5\%$, $F_{\text{mkt}} 0.6750$); Group 3 used CAD/USD (spot 0.7200, $r_{\text{USD}} 5\%$, $r_{\text{CAD}} 3.0\%$, $F_{\text{mkt}} 0.7600$). Each group computed the IRP no-arbitrage forward rate for their pair, identified the mispricing, and calculated the CIA profit per \$1,000,000. All three pairs exhibited $F_{\text{mkt}} > F_{\text{IRP}}$, confirming a borrow-USD / invest-foreign strategy. Results were submitted individually for grading and displayed on the class live feed.

3. FX Forward Contracts

Forward contracts are the most widely used corporate FX hedging instrument. They are traded over-the-counter between a corporation and its bank counterparty.

Feature	Description
Market	OTC — negotiated directly between firm and bank; no exchange.
Customization	Any notional amount, any settlement date, any deliverable currency pair.
Cost	No upfront premium. The forward rate ($F = \text{IRP rate}$) embeds the cost.
Settlement	Physical delivery (actual exchange of currencies) or net cash settlement.
Counterparty risk	Both parties bear credit exposure to one another.

Exit	Illiquid — unwinding before maturity requires negotiating an offsetting contract.
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Corporate use cases: exporters with foreign-currency receivables sell the foreign currency forward to lock in USD proceeds; importers with foreign-currency payables buy the foreign currency forward to lock in USD costs; multinationals hedge portfolios of cash flows across multiple currencies and maturities simultaneously.

4. FX Futures Markets

FX futures are standardized forward contracts traded on organized exchanges, principally the CME Group (Chicago Mercantile Exchange). They serve the same economic function as forwards but differ significantly in mechanics.

Feature	FX Futures (Exchange-Traded)	FX Forwards (OTC)
Trading Venue	CME Group exchange	Over-the-counter (bank)
Contract Size	Standardized (e.g., €125,000 per EUR contract)	Fully customizable
Margin & Settlement	Initial margin; daily mark-to-market (variation margin)	No daily cash flows; settled at maturity
Liquidity	High — easy entry and exit; most positions closed before delivery	Low — unwinding requires new negotiation
Transparency	Regulated by CFTC; published prices	Private bilateral
Counterparty Risk	Cleared by exchange (CME Clearing)	Direct bilateral credit risk

Most actively traded FX futures pairs at the CME: EUR/USD, GBP/USD, JPY/USD, CAD/USD, AUD/USD, CHF/USD, and MXN/USD. Contracts are quoted as USD per unit of foreign currency.

5. Case Study: Caterpillar Inc. and AUD Exposure

Caterpillar (CAT), a global machinery manufacturer, sold mining equipment to an Australian company with payment due in three months: AUD 1,000,000. The current AUD/USD spot is 0.6500, so the receivable is worth \$650,000 today. The 3-month AUD futures price is 0.6400 (approximately equal to the IRP forward rate, reflecting $r_{\text{AUD}} \approx 4\%$ versus $r_{\text{USD}} \approx 5.25\%$). CAT faces the risk of AUD depreciation before receipt, which would reduce USD revenues.

WORKED EXAMPLE 2

Caterpillar: Hedge vs. No-Hedge Comparison

Scenario	AUD/USD at Maturity	Unhedged USD Revenue	Hedged USD Revenue (Short Futures @ 0.6400)	Futures P&L;
AUD Appreciates	0.6800	AUD 1M $\times$ 0.6800 = \$680,000	\$640,000 (always)	$(0.6400 - 0.6800) \times 1\text{M} = -\$40,000$
No Change	0.6500	AUD 1M $\times$ 0.6500 = \$650,000	\$640,000	$(0.6400 - 0.6500) \times 1\text{M} = -\$10,000$
AUD Depreciates	0.6000	AUD 1M $\times$ 0.6000 = \$600,000	\$640,000	$(0.6400 - 0.6000) \times 1\text{M} = +\$40,000$

The futures hedge locks in exactly \$640,000 in every scenario: physical revenue + futures P&L; = $0.6400 \times \text{AUD } 1,000,000 = \$640,000$, always. The hedge sacrifices the \$680,000 upside to guarantee against the \$600,000 downside. Hedging is about certainty, not profit maximization.

Three primary reasons firms hedge FX exposure: (1) Certainty — lock in cash flows for reliable budgeting and financial planning; (2) Focus — remove FX noise from core business P&L;, so management attention stays on operations; (3) Stakeholders — reduce earnings volatility, which investors and lenders value independently of the expected hedge payoff.

IN-CLASS ACTIVITY · NovaTech Inc. — Dual-Currency Hedge Exercise

NovaTech, a U.S. technology firm, simultaneously faces two FX exposures over six months: a EUR receivable from a European client and a GBP payable to a UK supplier. Market data: EUR/USD spot = 1.0850, GBP/USD spot = 1.2700, $r_{\text{USD}} = 5.25\%$, $r_{\text{EUR}} = 3.75\%$, $r_{\text{GBP}} = 5.00\%$, $T = 0.5$ yr. Three student groups computed the six-month IRP forward rates for both pairs ($F_{\text{EUR/USD}} = 1.0930$ and $F_{\text{GBP/USD}} = 1.2716$) and then calculated their group's net USD position under the forward hedge. Group exposures: Group 1 — receive €500K, pay £300K; Group 2 — receive €750K, pay £200K; Group 3 — receive €600K, pay £400K. The exercise illustrated how firms manage portfolios of FX exposures — hedging each leg separately using IRP-implied forward rates.